

Report of the COMETA Workshop on Polarization Tagging for Hadronic Boosted Bosons

Perugia, Italy, 1–2 July 2026

Workshop summary

July 6, 2026

Abstract

This report summarizes the COMETA (COST Action CA22130) workshop on polarization tagging for hadronic boosted electroweak bosons, held in Perugia on 1–2 July 2026. The workshop brought together theorists, experimentalists, and machine-learning experts to discuss the theoretical, experimental, and ML-related aspects of boosted hadronic decays of W and Z bosons, with a particular focus on polarization tagging and vector-boson-scattering (VBS) topologies. The report is organized following the blocks of the workshop agenda: semi-leptonic decay channels, jet substructure and tagging, quantum machine learning, open-data discussion, Monte Carlo modeling of polarization, the experimental status of polarization measurements, machine-learning approaches, and the closing discussion towards a COMETA deliverable.

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1 Introduction and welcome

The workshop, organized by the COMETA COST Action (Comprehensive Multiboson Experiment-Theory Action, CA22130), took place at the Aula Magna of the Agraria Department of the Università di Perugia. In the opening session, Valentina Mariani (Università e INFN, Perugia) welcomed the participants and presented the practical organization and the goals of the meeting: to bring together the theoretical, experimental, and machine-learning communities working on boosted hadronic decays of electroweak bosons, with a spin on polarization tagging and on vector-boson-fusion/scattering topologies.

The physics case underlying the whole workshop can be stated concisely. Longitudinally polarized W and Z bosons are the physical remnants of the Higgs mechanism; in the Standard Model (SM) the scattering of longitudinal bosons ($V_L V_L \rightarrow V_L V_L$) is unitarized by delicate gauge and Higgs cancellations, so measuring polarized (in particular doubly longitudinal) diboson production and scattering is a direct probe of electroweak symmetry breaking (EWSB) and a sensitive portal to new physics. All measurements to date use fully leptonic decays, which are clean but statistically starved: the longitudinal fraction in VBS is only $\mathcal{O}(5\text{--}10)\%$ of an already rare process. Hadronic boson decays offer an order-of-magnitude larger branching fraction and full kinematic reconstruction in the boosted regime, where the decay is contained in a single large-radius jet, but require new tools—polarization-aware jet taggers, decorrelation techniques, precise Monte Carlo modeling of polarized production and hadronic decays—which formed the core of the two-day program.

2 Semi-leptonic decay channels (Day 1, morning)

2.1 Experimental review for diboson and VBS

Ankita Mehta (*Ghent University, for the ATLAS and CMS Collaborations*)

The session opened with a comprehensive review of the LHC electroweak multiboson program at the precision and energy frontiers. Diboson production (WW , WZ , ZZ , $V\gamma$) has been measured inclusively and differentially at 5.02, 13, and 13.6 TeV, with precision that in many areas already exceeds the original projections, thanks to detector performance, advanced analysis techniques, and higher-order theory (up to NNLO QCD + NLO EW). Highlights included the CMS anomalous-coupling analysis of semi-leptonic WV with ParticleNet-tagged boosted bosons (currently among the strongest bounds on triple-gauge and four-fermion operators), the ATLAS $W\gamma$ measurement with radiation-amplitude-zero and polarization-sensitive observables, and the ATLAS $ZZ \rightarrow 2\ell 2\nu(jj)$ cross sections with aTGC/aQGC constraints. On the VBS side, all major channels ($W^\pm W^\pm$, WZ , ZZ) are now observed; the first Run-3 VBS measurement (CMS $W^\pm W^\pm$ and WZ at 13.6 TeV, including first differential WZ distributions) is statistically limited, and semi-leptonic channels ($WVjj$, $ZVjj$, $VVjj$) increasingly rely on DNN/RNN-based signal-background discrimination. The speaker stressed that semi-leptonic modes are challenging but that ML-based boson tagging has been a game changer, that VBS simulations are still LO in many cases, and that combinations and SMEFT interpretations (including the ATLAS EW global fit) make these measurements key inputs to global searches for new physics.

2.2 Theory review on polarised bosons

Rene Poncelet (*IFJ PAN Kraków*)

The theory review traced the path from the motivation (unitarity of $W_L W_L$ scattering and its link to EWSB) to the state of the art in polarized predictions. Since boson polarization is not directly observable, decay products act as polarimeters; the modern approach defines polarized cross sections for on-shell bosons in the double-pole or narrow-width approximation (DPA/NWA), generating separately polarized samples that are fitted to data, with interference contributions kept under control. Higher-order corrections are essential: polarization fractions are shape-dependent and the corrections themselves depend on the polarization state, so unpolarized K -factors distort spectra; NNLO QCD is needed for percent-level scale uncertainties (illustrated with polarized W +jet at NNLO and mock-data fits). The talk reviewed the rapidly growing literature of polarized calculations—NLO QCD/EW for WW , WZ , ZZ , NNLO QCD for WW and (new) NNLO QCD+NLO EW for WZ with theory-nuisance-parameter uncertainties, semi-leptonic decays at NLO QCD (resolved and boosted setups, where LL turns out to be rather insensitive to QCD corrections while TT /mixed states receive large ones)—as well as Monte Carlo developments (polarized MadGraph, SHERPA nLO+PS, POWHEG-BOX-RES NLO+PS, and SMEFT@NLO+PS). A dedicated COMETA study of polarized ZZ production was presented, comparing NWA/DPA/off-shell treatments, matching/merging schemes, and best predictions against the ATLAS polarization fractions (agreement, dominated by experimental uncertainties). The outlook identified hadronic polarization taggers, higher-order corrections to decays, and parton-shower matching as the next frontiers.

2.3 Gauge boson polarization as a probe of SMEFT effects in WZ production (remote)

Maryam Bayat Makou (*Southern Methodist University; with S. Bhattacharya*)

This contribution presented a feasibility study, carried out within COMETA, of the interplay between dimension-6 SMEFT operators and gauge boson polarization in $WZ \rightarrow 3\ell$ production at $\sqrt{s} = 13$ TeV. Dedicated polarized samples ($W_T Z_T$, $W_L Z_L$, $W_T Z_L$, $W_L Z_T$; 100k events each, MadGraph + Pythia + Delphes) were compared between SM and SMEFT using a broad set of

kinematic and angular observables. EFT effects grow in the high- p_T tails (clearly visible in the leading-lepton p_T above ~ 200 GeV, dominated by the energy-growing behavior of C_W); ranking the operators by a polarization-shape metric shows that c_{HWB} , c_{HDD} and c_{H13} produce the largest genuine distortions of the polarization composition, while others (e.g. c_{HWB} at large coefficient) act mainly on the overall rate. A DNN trained on SM polarized samples to separate longitudinal from transverse configurations learns physically meaningful observables (the top feature being the lepton rapidity separation Δy), and EFT events systematically shift the learned polarization scores, providing operator-dependent “polarization fingerprints”. A two-output architecture predicting $P(W_L)$ and $P(Z_L)$ separately was advocated as more physical and better suited for EFT studies. A publication is in preparation.

2.4 Discussion

The session closed with a discussion on the complementarity between polarization observables and traditional EFT analyses, on the treatment of interference and off-shell effects in polarized templates, and on the priorities for extending these studies to semi-leptonic final states.

3 Jet substructure and tagging (Day 1, late morning)

3.1 Traditional jet-substructure observables

Lorenzo Mai (*Università di Genova & INFN Genova*)

This talk reviewed the “classical” toolbox of jet substructure for boosted boson identification, setting the baseline against which the ML and quantum-ML approaches of the following sessions are compared: large-radius jet reconstruction and grooming (trimming, soft drop), the groomed jet mass, prong-sensitive observables such as N -subjettiness ratios (τ_{21}) and energy-correlation functions (D_2), and their sensitivity—and limitations—when used to separate W/Z -initiated jets from the QCD background and, more ambitiously, longitudinal from transverse polarization states through the momentum sharing of the two subjets.¹ The discussion emphasized that high-level, IRC-safe substructure observables remain valuable because of their smaller sensitivity to parton-shower and hadronization modeling compared to low-level constituent-based taggers.

3.2 Machine learning for jet tagging (cancelled)

The scheduled overview by Sophia Vent on machine learning for jet tagging was cancelled; the corresponding material was partially covered by the flavour tagging review below and by the tagger-specific talks on Day 2.

3.3 Flavour tagging in ATLAS and CMS with machine learning

Alessandro Calandri (*Università & INFN Firenze, for ATLAS and CMS*)

The review traced the evolution of heavy-flavour tagging from multivariate combinations of impact-parameter, secondary-vertex, and soft-lepton information (MV2c10, DeepCSV) to the Run-2 paradigm shift of constituent-based deep learning (DeepJet) and today’s Transformer era. CMS Run-3 tagging is based on ParticleNet and Particle Transformer derivatives, culminating in UP-arT v2, a single unified model performing b/c /strange/quark-gluon/tau identification together with flavour-aware energy regression and resolution estimation, with adversarial training for robustness. ATLAS deploys GN2 and now GN3, with extended inputs (tracks, electrons, muons, particle-flow candidates) and auxiliary tasks (vertexing, track origin, heavy-quark charge), giving

¹The uploaded slide deck exceeded the size that could be retrieved for this summary (LM_UJ_2026.pdf); this paragraph is therefore based on the talk’s title, session context, and the discussion it generated, and is intentionally kept at a higher level than the other summaries.

substantial gains in c- and light-jet rejection, and ITk studies target the HL-LHC (improved pixel granularity and forward coverage relevant for VBF/VBS). Online b-tagging with ParticleNet at the CMS trigger and first c-tag triggers significantly enlarge the acceptance for $HH \rightarrow 4b$, $t\bar{t}H$, and boosted $H \rightarrow b\bar{b}/c\bar{c}$. Calibration remains the key to physics use: fixed-working-point and pseudo-continuous 2D b/c calibrations, optimal-transport (normalizing-flow) morphing of simulation to data in ATLAS, and dedicated calibrations for boosted-Higgs taggers (CMS GloParT with 314 classes, ATLAS GN2X) using $g \rightarrow b\bar{b}/c\bar{c}$ proxies and boosted $Z \rightarrow b\bar{b}$. Both experiments observe logarithmic scaling-law improvements with training-set and model size (ATLAS up to 7.7 billion jets and 86M parameters), suggesting performance has not yet saturated.

4 Quantum machine learning (Day 1, afternoon)

4.1 An introduction to quantum computing approaches for optimization problems

Marco Baiocchi (*Università degli Studi di Perugia*)

This pedagogical lecture introduced the two main quantum-computing paradigms relevant for optimization: gate-based programmable devices and quantum annealers. After reviewing qubits, superposition, entanglement, measurement, elementary gates, and circuit execution on NISQ (noisy intermediate-scale quantum) hardware, the speaker presented variational quantum algorithms, in which a parametric circuit is optimized by a classical outer loop, with QAOA (alternating problem and mixer Hamiltonians) as the prototypical algorithm for combinatorial problems—illustrated with charged-particle track reconstruction formulated as a QUBO over triplet-selection variables—and VQE for Hamiltonian ground-state problems. The second part covered adiabatic quantum computation and D-Wave-type annealers: the adiabatic theorem, annealing schedules, quantum tunneling as a mechanism to escape local minima, the QUBO formulation (with minimum vertex cover as a worked example), and practical knobs such as number of reads, annealing time, and reverse annealing for hybrid quantum–classical refinement. The take-away was that variational algorithms attempt to work around NISQ limitations on gate-based machines, while annealers are a pragmatic alternative for mid-size optimization problems.

4.2 Quantum-Informed Neural Networks

Aritra Bal, Michael Binder (*KIT; with M. Klute, B. Maier, M. Spannowsky*)

Rather than running full quantum classifiers, this work extracts geometric information from quantum representations of jets and injects it into classical networks. Jets are encoded with the one-particle-one-qubit (1P1Q) scheme (up to ten constituents, (p_T, η, ϕ) per qubit); the Quantum Fisher Information Matrix (QFIM) of the resulting state is interpreted as a learned summary of two-particle correlations: for top jets, large off-diagonal QFIM blocks connect subjet constituents, while QCD jets show suppressed off-diagonals. Three increasingly powerful applications were shown: (i) a training-free, QFIM-only prototype classifier reaching VQC-level performance with faster, more stable convergence when used as initialization; (ii) a non-trainable graph classifier and a trainable GNN (“QuantumConv1D”) whose messages are augmented with QFI sub-matrices, converging faster and outperforming the classical benchmark; and (iii) QUIVER, which injects QFI tokens and attention biases into the state-of-the-art Particle Transformer, giving statistically significant improvements over five runs on top tagging and $H \rightarrow c\bar{c}$ benchmarks (arXiv:2606.02785, accepted at the ICML 2026 AI4Physics workshop). Current limitations are the ~ 10 -qubit simulability ceiling and the cost of QFIM-based optimization; the outlook includes extending from top to boosted $W/Z/H$ jets—directly relevant to polarization tagging—and hybrid quantum-to-classical pipelines beyond qubits.

4.3 Quantum Lund plane for VBS

Fabrizio Napolitano (*Università e INFN Perugia*)

This talk (based on arXiv:2604.18613) proposed a quantum machine-learning approach to boosted-boson and polarization tagging built on the Lund jet plane. The LP2B (Lund Plane to Bloch) encoding maps each Lund declustering node, through a differentiable and trainable stereographic projection, onto a qubit of a Quantum Tree Tensor Network (QTTN) whose topology mirrors the C/A declustering tree (kept shallow via a $\ln k_t > 1$ cut), with CRY entanglement, variational layers, and readout; by construction the representation is IRC-safe. Benchmarked against BDTs and MLPs on Lund inputs, LundNet-2, and 1P1Q-style VQCs, on tasks including W tagging, top tagging, and W /graviton polarization discrimination, the QTTN reaches competitive performance with a very small parameter count and no sign of overfitting even with only 10^3 training jets. A central physics message was that polarization tagging is “difficulty-limited” (W_T and W_L are the same particle, so methods saturate quickly), and that low-level constituent-based approaches inherit parton-shower/hadronization modeling uncertainties of up to $\sim 40\%$, whereas IRC-safe high-level (Lund-plane) inputs are far more resilient—a decisive consideration for analyses with large backgrounds in the signal region. The model was validated on real quantum hardware (SpinQ Triangulum, 3 qubits), with AUC compatible with simulation (0.58 ± 0.02 vs 0.60 ± 0.02).

4.4 Discussion

The session discussion focused on where quantum approaches can be genuinely complementary to classical ML—compact, physics-informed representations and modeling robustness rather than raw performance—and on the scalability of these methods towards realistic detector-level datasets.

5 Discussion session: open data (Day 1, late afternoon)

5.1 Open data proposal

Karolos Potamianos (*University of Warwick*)

This contribution (no slides were posted on Indico) put forward a concrete proposal, in the spirit of community datasets such as JetClass and the CMS Open Data used by several speakers during the day, for a shared, well-documented open dataset tailored to polarization tagging of hadronic boosted bosons: common polarized signal samples (W_L/W_T , Z_L/Z_T) and QCD backgrounds with agreed generator settings, truth-labeling conventions, and evaluation metrics, so that the taggers presented at this workshop (ParT-based, Lund-plane/QTTN, DDT-decorrelated, regression-based) can be compared on equal footing and reused across ATLAS, CMS, and the phenomenology community.

5.2 Open data roundtable

Antonio Giannini (USTC), **Nurfikri Norjoharuddeen** (LUT) (*conveners*)

The roundtable gathered experimental and theory input on the practical aspects of the proposal: which level of information to release (parton, particle, Delphes-like, or full-simulation constituents), how to encode polarization definitions consistently (frame choices, DPA/NWA conventions, treatment of interference) so that samples remain theoretically meaningful, licensing and hosting (Zenodo/CERN Open Data), and how a common benchmark could feed directly into the COMETA deliverable discussed on Day 2. There was broad agreement that a public, polarization-labeled boosted-boson dataset would significantly lower the entry barrier for new groups and make performance claims comparable across the field.

6 MC modeling, including polarisation (Day 2, morning)

6.1 Measuring the polarization of boosted, hadronic W bosons with jet substructure observables

Songshaptak De (*Jožef Stefan Institute; with V. Rentala and W. Shepherd, JHEP 05 (2025) 028*)

A new, production-process-independent technique to extract the longitudinal and transverse polarization fractions of boosted hadronic W bosons was presented. The key object is a jet-substructure observable $p_\theta = \Delta E^{\text{reco}}/p_W^{\text{reco}}$, the subjet energy difference normalized to the fat-jet momentum, which at parton level equals $|\cos\theta^*|$, the decay polar angle in the W rest frame; it is shown to have smaller reconstruction errors than the traditionally used opening-angle proxy, especially at large boosts, and remains faithful up to $|\cos\theta^*| \lesssim 0.9$. Hadron-level 2D templates in (p_θ, τ_{21}) are built for longitudinal and transverse W jets (scalar/pseudoscalar $\phi \rightarrow WW \rightarrow 4q$ samples, $800 < p_T < 1000$ GeV, soft-drop groomed anti- k_t $R = 1.0$ jets) and for the QCD dijet background, and fitted to pseudo-data. Applied to semi-leptonic WW VBS at the HL-LHC, the method reconstructs the *transverse* fraction to 20–30% with 3–10 ab^{-1} , but with SM production alone the *longitudinal* fraction cannot be distinguished from zero even with 10 ab^{-1} ($f_L = 0.0075 \pm 0.0045$ for a true 0.0074), because of the $\mathcal{O}(25:1)$ QCD background and the tiny LL rate. The technique would however be sensitive to BSM-enhanced longitudinal production, and can be strengthened by using p_θ inside a multivariate analysis.

6.2 A systematic approach to amplitude-assisted polarization tagging (remote)

Erik Bachmann (*TU Dresden; with D. M. Hoppe*)

This theory contribution addressed the question of what the *optimal* polarization observable is. In contrast to classification-based strategies, which approximate polarization fractions through relative frequencies of purely polarized training samples, amplitude-assisted polarization tagging learns per-event polarization fractions directly from the matrix elements. The talk presented a systematic construction of the corresponding optimal observable, which defines the fundamental limit for separating polarization components and therefore provides a natural benchmark against which neural-network approaches can be tested—enabling a diagnosis of where and why their performance falls short of optimality. The methodology was illustrated on the simplest non-trivial process, leptonic Z +jet production at leading order, and extensions to W and diboson processes as well as to hadronic decay channels were discussed.

6.3 MC roundtable and discussion

Emanuele Re, Giovanni Pelliccioli (*Università & INFN Milano-Bicocca, conveners*)

The roundtable, continuing the thread opened by the Day-1 theory review, discussed the status and interoperability of polarized event generation: polarized samples in MadGraph (LO and multi-jet merged), SHERPA (nLO+PS), POWHEG-BOX-RES (NLO+PS, and now SMEFT@NLO+PS), and fixed-order DPA computations (MoCaNLO, NNLO QCD frameworks). Central topics were the definition dependence of polarized signals (frame choices, NWA vs DPA vs off-shell, interference and Breit–Wigner treatments), the size of higher-order corrections to hadronic decays and their interplay with parton-shower matching in the boosted regime, and the recommendation that experiments quote results with clearly documented polarization conventions to allow reinterpretation. The ongoing COMETA comparative study of polarized ZZ (and its planned extensions to WZ and to semi-leptonic final states) was identified as the natural vehicle to converge on best practices and best predictions.

7 Experimental status on polarisation (Day 2, late morning)

7.1 VBS and diboson polarisation measurements

Joany Manjarres (*Université de Toulouse, CNRS/IN2P3, L2IT*)

This review presented the full landscape of polarization measurements, from LEP (where longitudinal-longitudinal joint polarization was never observed) to the LHC. The formalism was set up in terms of the single- and two-boson spin density matrices, whose diagonal elements give the helicity fractions (f_{LL} , f_{LT} , f_{TL} , f_{TT}) and whose off-diagonal elements encode quantum coherence and entanglement. Measurements proceed by fitting data with polarized templates, whose construction is itself a challenge: LO MadGraph templates can bias fractions by 10–50%, so NLO-accurate templates are obtained by DNN-based multi-dimensional reweighting to fixed-order DPA predictions. The milestones reviewed were: CMS and ATLAS single- and joint-polarization in WZ (first observation of joint states, f_{00} at 7.1σ); the ATLAS high- $p_T(Z)$ analysis observing the radiation-amplitude-zero dip through a dedicated “depth” variable and reaching observation/evidence of doubly longitudinal bosons in the Goldstone-like regime; ATLAS evidence for $Z_L Z_L$ in $ZZ \rightarrow 4\ell$ (4.3σ); the first VBS-phase-space polarization measurements in same-sign WW (CMS, and the 2025 ATLAS evidence for $W_L W_X$ at 3.3σ with limits on $W_L W_L$); and the brand-new ATLAS and CMS $H \rightarrow ZZ^* \rightarrow 4\ell$ polarization analyses, which access off-diagonal density-matrix elements and reject the non-entangled pure-longitudinal hypothesis at 4.7σ (ATLAS) and $> 6\sigma$ (CMS)—the first entanglement tests with massive vector bosons. Concluding remarks: results should be presented in harmonized language (fractions vs pseudo-cross-sections vs density-matrix coefficients); $q\bar{q}$ - and gg -initiated production may deserve separate treatment; leptonic channels will remain statistics-limited even at the HL-LHC, so exploiting hadronic decays with boson taggers and matrix-element regressions is the way forward.

7.2 A polarisation-aware classification and regression tagger for boson-initiated jets (ATLAS)

Antonio Giannini (*USTC, based on ATLAS public plots JETM-2026-01*)

The first public ATLAS results on a polarization-aware extension of boosted boson tagging were shown. A Particle Transformer trained on UFO large- R soft-drop jet constituents performs a five-class classification (W_L , W_T , Z_L , Z_T , quark/gluon) augmented with regression auxiliary tasks ($\cos\theta^*$ and the kinematics of the two truth subjets). Key findings: quark/gluon rejection depends on the boson polarization state; the L -vs- T task is intrinsically hard (the classes overlap physically—a longitudinal boson at high $\cos\theta^*$ looks exactly like a transverse one) and the ParT discriminant already sits close to the theoretical upper bound, outperforming subjet-reclustering and quark-level $\cos\theta^*$ baselines. The more impactful novelty is the regression: the regressed $\cos\theta^*$ has the same response for L and T jets, achieves an absolute resolution of $\mathcal{O}(10\text{--}15\%)$ —up to a factor two better than reclustered subjets at low $\cos\theta^*$, where detector granularity forms a “wall”—and provides, for the first time, experimental observables directly sensitive to polarization in hadronic decays. Two-prong kinematic regression outperforms reconstructed subjets as well. A first Run-3 data/MC validation in a dijet- and V +jets-enriched selection (13.6 TeV, 28 fb^{-1} , inference via the ATLAS BoostedJetTaggers/ONNX infrastructure) shows well-behaved discriminant and regressed- $\cos\theta^*$ distributions.

7.3 Experimental perspectives in polarisation tagging

Antonio Giannini (*USTC*)

A companion big-picture talk addressed the Why/How/What/When of hadronic polarization measurements. Why: high-energy $V_L V_L$ interactions are the directest test of unitarization by the Higgs sector and a prime hiding place for new physics; all current polarization measurements are leptonic, while hadronic channels uniquely open the high- p_T regime. How: today’s boson taggers

solve V -vs- q/g classification and are calibrated on SM candles; polarization adds the harder L -vs- T problem, where jet-substructure BDTs leave $\sim 33\%$ contamination at 50% efficiency and deep learning on constituents or energy correlators only marginally improves—motivating the classification-plus-regression strategy of the previous talk, together with dedicated calibration schemes. What: inclusive V/VV production (potentially up to boson $p_T \sim 1$ TeV) can serve as measurement and calibration ground; the VBS semi-leptonic channel, already observed in Run 2 with a significant boosted contribution, is the flagship target; EFT constraints (dim-6/dim-8) depend on polarization states and would benefit directly. When: hadronic analyses historically lag because of tagger calibration, so realistic new measurements are expected on the full Run-2+3 datasets and at the HL-LHC; the pheno groundwork presented at this workshop (Rej, Napolitano, Allene, this tagger) is the necessary first step.

7.4 Discussion

The session discussion revolved around calibration strategies for a polarization-aware tagger (which SM candles can anchor an L/T -sensitive score?), the treatment of the model dependence introduced when polarization information enters the discriminant, and coordination between ATLAS and CMS on common conventions.

8 Machine-learning approaches (Day 2, afternoon)

8.1 Prospect studies for polarization measurements in boosted VBS semi-leptonic events

Tom Allene (*L2IT Toulouse; with A. Giannini and J. Manjarres*)

This prospect study targets the semi-leptonic VBS diboson final state as a benchmark for longitudinal-polarization measurements: the hadronic decay of one boson brings the large branching fraction and access to high boson p_T , while the leptonic boson retains a clean signature—but the longitudinal fraction in VBS is only $\mathcal{O}(7\%)$, making the measurement extremely challenging. The analysis is performed at truth level with MadGraph+Pythia simulations, with object selections approximating detector acceptance, and focuses on events where the hadronic boson is reconstructed as a boosted large-radius jet, in which polarization information is not directly accessible experimentally. Jet-level variables and jet constituents are used to recover polarization-sensitive observables (in the spirit of the regression approach shown in the ATLAS tagger talk), and the sensitivity to longitudinal polarization of a baseline cut-based analysis using classical jet and lepton observables is compared with that of an analysis exploiting ML-driven regression of polarization-sensitive variables, quantifying the gain that such techniques can bring to future ATLAS semi-leptonic VBS polarization measurements.²

8.2 Unlocking hadronic W/Z polarization measurements via a 3D Designed Decorrelated Tagger

Amartya Rej (*TU Dortmund*)

This contribution argued for a change of strategy in how taggers are used for polarization: a discriminant trained to separate L from T learns exactly the angular correlations one wants to measure, biasing any subsequent shape analysis. Instead, one should use a QCD-vs-signal tagger explicitly *decorrelated* from the polarization-sensitive observable X (a $\cos\theta^*$ proxy built from declustered subjet momenta in the large- R jet rest frame), preserving the angular information for unbiased extraction. Concretely, the talk presented a three-dimensional extension of the Designed Decorrelated Tagger (DDT, JHEP 05 (2016) 156) that simultaneously decorrelates an existing

²The uploaded slides ([Perugia_Presentation.pdf](#)) exceeded the retrievable size; this summary follows the contribution abstract and the discussion.

tagger score from the soft-drop mass, the jet p_T , and X , guaranteeing a flat QCD selection efficiency at a fixed working point and thereby enabling data-driven background estimation in bins of X . Using the public JetClass dataset with ParT and ParticleNet scores, the 3D DDT was shown to successfully flatten the QCD response versus X with minimal signal-efficiency loss; several X definitions were implemented and systematically ranked by decorrelation quality. The result is a concrete, tagger-agnostic analysis blueprint for hadronic W/Z polarization measurements at the LHC and HL-LHC.

8.3 Discussion

The discussion contrasted the two philosophies presented across the workshop—polarization-aware taggers plus regression versus polarization-blind, decorrelated taggers plus template fits in unbiased observables—and converged on the view that they are complementary: the former maximizes per-event information, the latter maximizes robustness of the background estimate, and both should be exercised on the proposed common open dataset.

9 Closing discussion (Day 2, late afternoon)

9.1 Roundtable on Monte Carlo

The final MC roundtable consolidated the actions identified on the Day-2 morning session: extend the COMETA polarized- ZZ comparison to WZ and to semi-leptonic/boosted final states; converge on documented conventions for polarized signal definitions (frames, DPA/NWA, interference) to be used by ATLAS, CMS, and the generator authors alike; and quantify parton-shower and hadronization uncertainties specifically for polarization-sensitive substructure observables, where low-level taggers were shown to be most vulnerable.

9.2 Towards a COMETA deliverable

The workshop closed with a planning session on the COMETA deliverable that should crystallize the outcome of the meeting. The envisaged product combines: (i) a common, public, polarization-labeled dataset of boosted hadronic W/Z jets and QCD background (following the open-data proposal of Day 1), with agreed truth definitions grounded in the MC roundtable conventions; (ii) a benchmark comparison of the tagging strategies presented—traditional substructure, Particle-Transformer classification and regression, quantum Lund-plane networks, decorrelated (3D DDT) taggers, and amplitude-assisted optimal observables—with common metrics including modeling robustness; and (iii) a road map for the first hadronic/semi-leptonic polarization measurements on the Run-2+3 datasets and their HL-LHC projections. A written COMETA report collecting these elements, with contributions from the workshop speakers, was agreed as the concrete next step.

10 Proposal: COMETA-POLJETCLASS, an extended JetClass dataset with polarization labels

Building on the open-data discussion of Day 1 and on the deliverable planning of Day 2, the participants converged on a concrete joint project: the construction of an extended, *polarization-labeled* JetClass-like dataset for semi-leptonic diboson processes, together with a community benchmark of the tagging architectures presented at this workshop. The project deliberately follows the blueprint of the Particle Transformer paper (Qu, Li, Qian, arXiv:2202.03772, ICML 2022), which paired a new large-scale public dataset (JetClass: 100M jets in 10 classes, generated with MadGraph+Pythia and a Delphes CMS-like detector, anti- k_t $R = 0.8$ jets with $p_T \in [500, 1000]$ GeV

and per-particle kinematic, identification, and trajectory-displacement features) with a new architecture (ParT, with pairwise interaction features in the attention bias) and a standardized evaluation protocol. The sections below mirror that structure.

10.1 The COMETA-POLJETCLASS dataset

Processes and classes. The dataset targets semi-leptonic ZV production, $V = W, Z$ decaying hadronically: $pp \rightarrow Z(\rightarrow \ell\ell) W(\rightarrow q\bar{q}')$ and $pp \rightarrow Z(\rightarrow \ell\ell) Z(\rightarrow q\bar{q})$, in both QCD-induced and VBS/EW ($ZVjj$) phase spaces, with the hadronically decaying boson boosted and captured by a single large-radius jet. The primary jet classes are $\{W_L, W_T, Z_L, Z_T, q/g\}$, the light quark/gluon class being drawn from $Z(\rightarrow \ell\ell)$ +jets to match the signal topology. The leptonic Z leg is retained in the event record so that *production-level* observables (scattering angle, system kinematics) are available alongside the jet, unlike in JetClass, which stores isolated jets only.

Improved MC accuracy. The distinguishing ambition, motivated by the theory review and the MC roundtable, is generation accuracy beyond the LO-template standard, in two directions:

1. *Perturbative accuracy.* Polarized signals are defined in the double-pole approximation with documented frame conventions, generated at NLO QCD accuracy (SHERPA nLO+PS and POWHEG-BOX-RES NLO+PS), supplemented by LO multi-jet-merged MadGraph samples; fixed-order DPA predictions (NLO and, where available, NNLO QCD) are used to validate or reweight the samples. This addresses the 10–50% biases of LO polarized templates highlighted in the experimental review.
2. *Multiplicity of generators and showers.* Every process/class is produced in several generator $\times$ shower combinations (at minimum MadGraph+Pythia8, SHERPA native, POWHEG+Pythia8, POWHEG+Herwig), spanning dipole and angular-ordered showers, with varied hadronization models), and each event carries a provenance label. Modeling robustness thus becomes a first-class, *measurable* property of any tagger trained on the dataset, directly probing the up-to-~40% shower/hadronization vulnerability of low-level taggers reported at the workshop.

Detector emulation, features, and labels. For comparability with JetClass, events are passed through the same Delphes CMS-like card, with anti- k_t $R = 0.8$ jets in $p_T \in [400, 1000]$ GeV, and per-particle features form a strict superset of the JetClass ones (kinematics, particle identification, trajectory displacement). Each jet additionally carries truth labels: the boson helicity state in the declared frame, the true decay angle $\cos\theta^*$ (and azimuth φ^*), the production/scattering angle Θ of the diboson system, the truth subjet four-momenta, and the boson p_T . A target size of $\mathcal{O}(10)$ M jets per class per generator/shower combination keeps the total within an order of magnitude of JetClass while enabling scaling-law studies.

10.2 Architectures to be compared

The benchmark will confront, on identical inputs and splits, the families of approaches presented at the workshop: (i) a high-level, IRC-safe baseline: BDTs/MLPs on groomed mass, τ_{21} , D_2 , energy correlators, and the p_θ observable of the De-Rentala-Shepherd method; (ii) constituent-based deep networks: ParticleNet, a plain Transformer, and ParT with its pairwise-interaction attention bias, in the five-class configuration and in the ATLAS-style multi-task configuration with $\cos\theta^*$ and subjet-kinematics regression heads; (iii) Lund-plane approaches: LundNet as classical reference and the LP2B quantum tree tensor network, testing whether IRC-safe declustering inputs retain their robustness advantage at scale; (iv) quantum-informed hybrids: QUIVER-style injection of quantum Fisher information tokens into ParT; (v) decorrelated variants: the 3D DDT applied to the best classifier scores, flattening the QCD response in mass, p_T , and the polarization-sensitive observable; and (vi) the amplitude-assisted optimal observable constructed from matrix elements, serving as the per-event information upper bound against which all learned approaches are gauged.

10.3 Benchmark tasks and metrics

Following the ParT evaluation protocol, but extended to the polarization context, four tasks are defined:

1. *Tagging*: each polarized boson class versus q/g , reported as accuracy, AUC, and background rejection at 50% and 99% signal efficiency, *separately per polarization state*, since the workshop showed that q/g rejection depends on polarization.
2. *Polarization classification*: V_L versus V_T (and the joint W -vs- $Z \times L$ -vs- T problem), with AUC complemented by a physics-level figure of merit: the expected uncertainty on the longitudinal fraction f_L from template pseudo-fits including the QCD background, which is what an LHC analysis ultimately optimizes.
3. *Regression*: per-jet estimation of $\cos\theta^*$ (and optionally φ^* and the subset kinematics), evaluated through resolution (σ_{68}), bias versus true $\cos\theta^*$, and—critically—response uniformity between L and T samples, so that the regressed observable can be used in an unbiased template fit.
4. *Decay versus production angles*: a dedicated study quantifying whether the decay angle $\cos\theta^*$ (jet-internal information) or the production angle Θ (event-level information, accessible thanks to the retained leptonic leg) carries more polarization sensitivity per event, using Fisher-information and mutual-information estimates as well as the degradation of the f_L pseudo-fit when either input is removed. This directly informs whether future analyses should invest in jet-substructure regression, in event-level reconstruction, or in their combination.

All results are to be reported with a robustness matrix (train on generator/shower A , test on B) and, for decorrelated taggers, with the sculpting of the QCD templates quantified by a Jensen–Shannon divergence.

10.4 Suggestions

The following suggestions were added by the author of this summary. (1) Store, for every event, the per-helicity DPA matrix-element weights (including the interference term): this makes the amplitude-assisted optimal observable computable on the released events, allows reweighting between polarization definitions and reference frames without regeneration, and turns the definition dependence itself into a quantifiable systematic. (2) Provide helicity labels in at least two frames (e.g. the VV center-of-mass helicity frame and the Collins–Soper frame), since polarization fractions are frame-dependent and ATLAS/CMS conventions differ. (3) Include an explicit “interference/off-shell” event category rather than forcing every event into a pure-helicity class, so benchmark fits can propagate the non-closure of the polarized decomposition. (4) Reweight all classes to a common boson- p_T spectrum (or provide the weights) to prevent taggers from learning production kinematics instead of decay structure—particularly important for the decay-vs-production-angle study of Task 4. (5) Publish fixed train/validation/test splits, evaluation code, and a public leaderboard in the style of the `jet-universe` repository, with a versioned DOI (Zenodo/CERN Open Data), and require metrics to be quoted as mean $\pm$ spread over at least five trainings. (6) Add a small full-simulation companion sample (ATLAS/CMS open data) to test ParT-style fine-tuning and quantify the Delphes-to-full-sim transfer gap before any in-experiment deployment. (7) Extend the ParT pairwise interaction features with polarization-motivated pair variables (e.g. the momentum-sharing z and pair mass in the jet rest frame), a cheap architectural handle that directly encodes the physics of the L/T decay distributions $\propto \sin^2\theta^*$ vs $(1 \pm \cos\theta^*)^2/2$. (8) Include the azimuthal decay angle φ^* among the regression targets from the start: it gives access to the off-diagonal spin-density-matrix elements and thus connects this dataset to the quantum-information program (entanglement observables) highlighted in the experimental review.

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